

DATA MINING & MACHINE LEARNING

[Comparison of ML models for classification of heart disease]

Data Mining & Machine Learning Project

M.Sc. in Artificial Intelligence and Data Engineering — Academic Year 2025–2026

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Presentation Outline

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Problem & Dataset

Task definition · features · EDA

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Pipeline & Experiments

Preprocessing · model selection · hyperparameter optimization

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Evaluation & Results

Metrics · cross-validation · final model performance

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Critical Discussion

Error analysis · limitations · future directions

PART 1

PART 2

PART 1 — TECHNICAL

Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

Problem Definition & Dataset

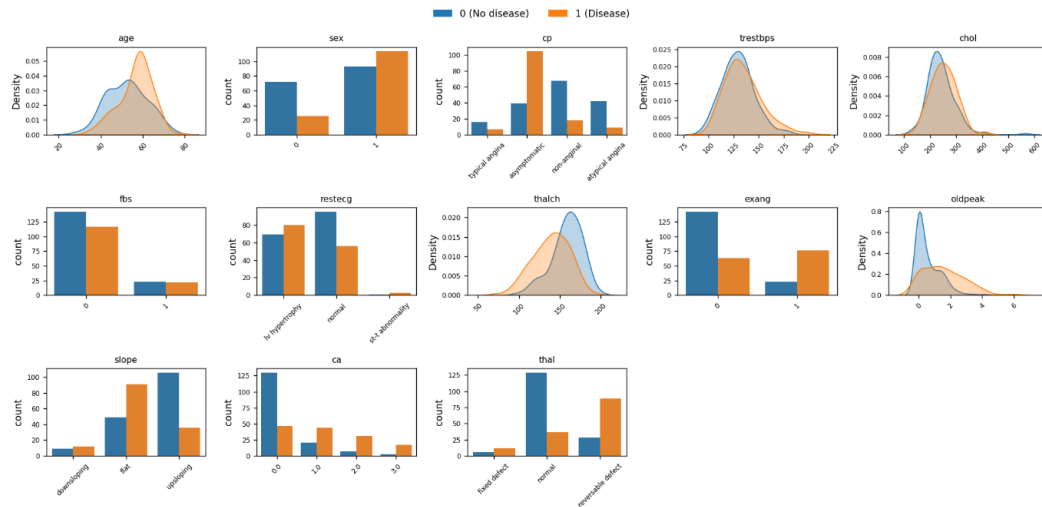
The Problem

- Compare 3 different classification models, 2 different dataset subsets and the effect of dataset balancing on the results.
- Task type: classification
- Heart disease is a leading cause of death worldwide. Early detection is very important to prevent it.
- Key challenge – dataset cleaning

Dataset at a Glance

Task	Binary classification
Dataset	Source: UCI Heart Disease, full – 920 samples, 10 features, Cleveland – 304 samples, 13 features
Target	Num – 0 = no heart disease, 1 = heart disease present
Features	6 numerical features, 8 categorical features
Missing values	Original: 11.95, Full: 5.20, Cleveland: 0.21
Class balance	Full: 1 = 55.32%, 0 = 44.67, Cleveland: 1 = 54.27, 0 = 45.72
Split	Evaluation: Stratified 5-Fold Cross-Validation
Metric(s)	Accuracy, Precision, Recall, F1-Score, ROC-AUC

Exploratory Data Analysis- Cleveland



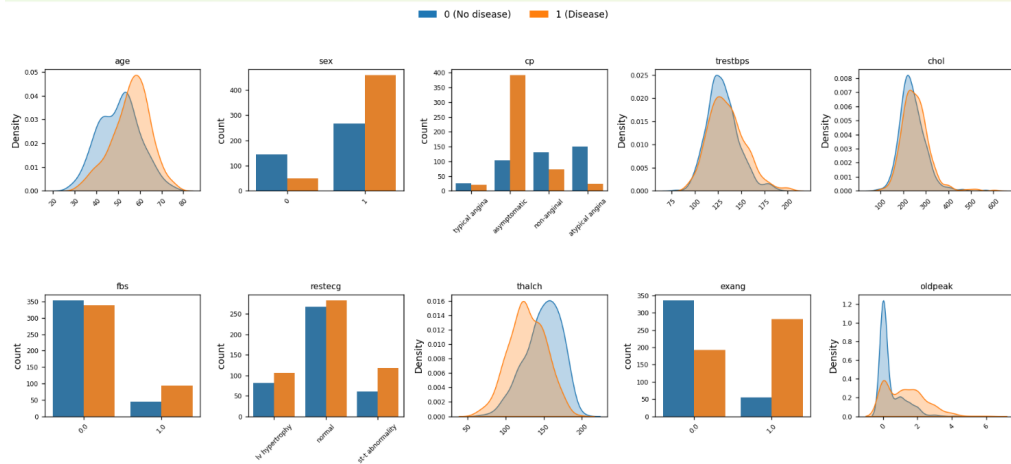
Cleveland Subset

	age	ca	cp	exang	oldpeak	sex	slope	thal	thalch	num
age	1.00	0.36	-0.19	0.10	0.21	-0.10	-0.12	0.10	-0.40	0.23
ca	0.36	1.00	-0.19	0.15	0.30	0.09	-0.11	0.16	-0.26	0.46
cp	-0.19	-0.19	1.00	-0.32	-0.34	-0.12	0.22	-0.17	0.28	-0.36
exang	0.10	0.15	-0.32	1.00	0.29	0.14	-0.24	0.21	-0.38	0.43
oldpeak	0.21	0.30	-0.34	0.29	1.00	0.10	-0.55	0.22	-0.35	0.43
sex	-0.10	0.09	-0.12	0.14	0.10	1.00	-0.05	0.20	-0.04	0.27
slope	-0.12	-0.11	0.22	-0.24	-0.55	-0.05	1.00	-0.05	0.36	-0.32
thal	0.10	0.16	-0.17	0.21	0.22	0.20	-0.05	1.00	-0.11	0.34
thalch	-0.40	-0.26	0.28	-0.38	-0.35	-0.04	0.36	-0.11	1.00	-0.42
num	0.23	0.46	-0.36	0.43	0.43	0.27	-0.32	0.34	-0.42	1.00

Key findings from EDA:

- Most correlated features – *ca, exang, oldpeak, thalch* (target = *num*)
- Classes mostly not distributed evenly
- Numerical variables mostly follow normal distribution but often skewed.
- Very little missing values, only present in *slope, ca* and *thal*.
- Class 0, on average, has more missing values than class 1.

Exploratory Data Analysis - Full



Full Subset

	age	cp	exang	oldpeak	sex	thalch	num
age	1.00	-0.21	0.20	0.26	0.06	-0.37	0.28
cp	-0.21	1.00	-0.33	-0.32	-0.18	0.28	-0.46
exang	0.20	-0.33	1.00	0.40	0.18	-0.36	0.46
oldpeak	0.26	-0.32	0.40	1.00	0.11	-0.16	0.40
sex	0.06	-0.18	0.18	0.11	1.00	-0.18	0.31
thalch	-0.37	0.28	-0.36	-0.16	-0.18	1.00	-0.39
num	0.28	-0.46	0.46	0.40	0.31	-0.39	1.00

Key findings from EDA:

- Most correlated features – cp, exang, oldpeak and thalch (target = num)
- Distribution similar to Cleveland subset, with small differences present.
- More missing values than Cleveland. Most missing values in chol, trestbps and fbs.
- Class 1, on average, has more missing values than class 0.

Feature Engineering & Selection

Engineering

- No engineered features.
- Winsorization to remove strong influence of outliers.
- Feature scaling to ensure compatible feature ranges.
- Categorical variables encoded using OneHotEncoder.
- Missing values imputed using most frequent class or median.
- Data transformations requiring statistics fitted only on training fold and applied to validation.

Selection Method

- All features were retained, due to the character of clinical data and potential non-linear relationships.
- Highest permutation importance for best performing model – ca.

ML Pipeline: Avoiding Data Leakage & Bias



Data Leakage Prevention

- All preprocessing requiring data statistics fitted only on training folds
- Data leakage safe operations done globally once to reduce overhead
- No information from val used during training

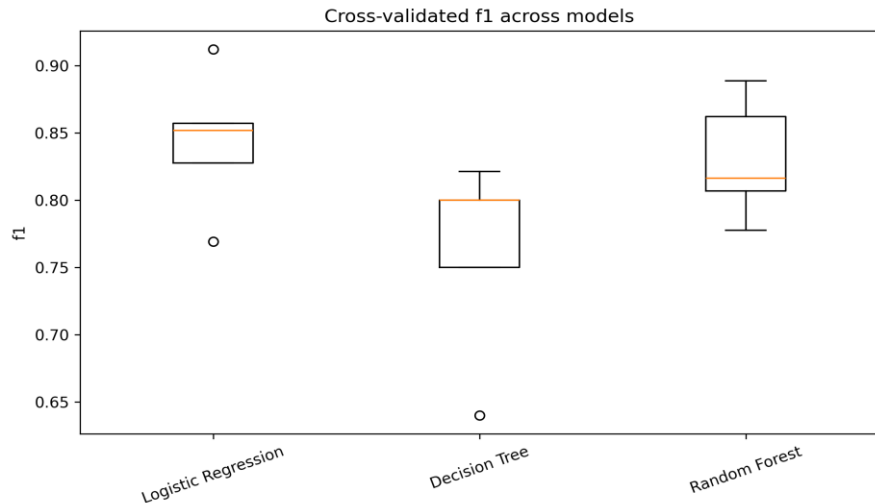
Class Imbalance Handling

- Class balancing used – comparison between not using it and using it
- Stratified k-fold validation to account for class imbalance
- Metrics chosen: Accuracy, Precision, Recall, F1 Score, ROC-AUC

Metric Justification

- Importance of FN in medical diagnosis – recall is particularly important
- ROC-AUC is less sensitive to class distribution and provides threshold independent measure of discriminative performance
- F1 score provides a balanced measure of comparison between models and works well with imbalanced datasets
- Precision importance for avoiding unnecessary follow-up tests
- Accuracy as a baseline metric

Model Selection & Hyperparameter Optimization



Key Decisions

Candidate models	Logistic Regression, Decision Tree, Random Forest
CV strategy	Stratified 5-Fold Cross-Validation
Selection criterion	Mean F1-score
Chosen model	Logistic Regression
Search strategy	Grid Search
Search space	Logistic Regression $C=\{0.01, 0.1, 1, 10, 100\}$, Decision tree: max_depth, samples_leaf, Random forest: n_estimators, max_depth, max_features, min_samples_leaf
Best config	Cleveland unbalanced: Logistic Regression $C=0.1$, Cleveland balanced: Logistic Regression $C=0.001$, Full unbalanced: Random Forest: 100 trees, depth=None, leaf=4, Full balanced: Random Forest 100 trees, depth=None, leaf=1
Statistical test	Wilcoxon signed-rank: $p=0.313$, Paired t-test: $p=0.251$

Dataset & Balancing Comparison

Comparison	Mean F1	Wilcoxon p	t-test p	Interpretation
Full RF unbalanced vs balanced	0.831 vs 0.824	0.125	0.041	Small advantage for unbalanced
Cleveland LR unbalanced vs balanced	0.844 vs 0.837	0.438	0.387	No significant difference
Cleveland LR vs Full RF	0.844 vs 0.831	0.625	0.680	No significant difference

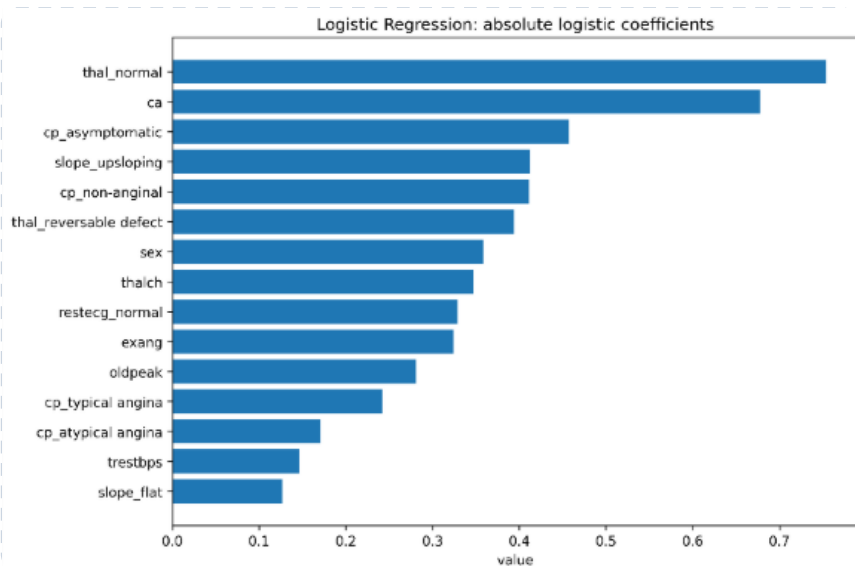
Statistical testing showed no significant differences between balanced and unbalanced variants or between the best Cleveland and Full configurations (Wilcoxon $p > 0.05$). However, consistent performance trends favored the Cleveland dataset and the unbalanced variants.

PART 2 — CRITICAL ANALYSIS

Explainability & Critical Discussion

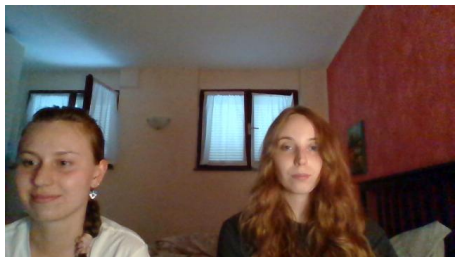
How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

Global Explainability: What Drives the Model Overall?



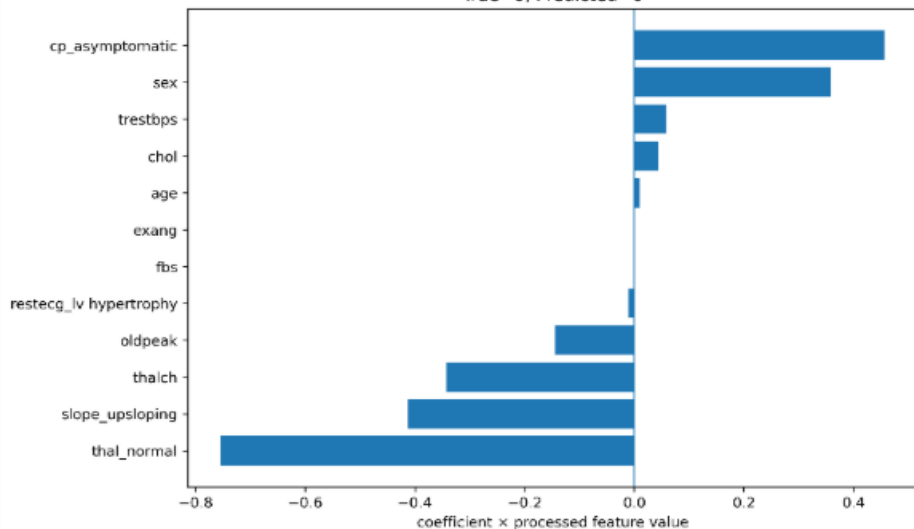
Interpretation

- Technique: Feature importance was evaluated using the absolute values of Logistic Regression coefficients. Larger coefficients indicate a stronger influence on heart disease prediction.
- Most important features: The most influential variables were `thal_normal`, `ca`, `cp_asymptomatic`, `exang`, and `oldpeak`. These features are known cardiovascular risk factors and are clinically meaningful.
- Interesting findings: The same features also appeared among the most important predictors in Random Forest, suggesting consistent results across different models.
- Does the global explanation support performance? Yes. The model relies on medically relevant variables rather than random patterns, which increases confidence in its predictions.
- Similar explainability patterns were observed across datasets and models.



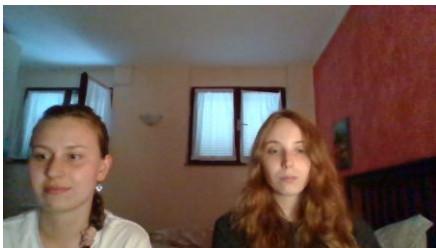
Local Explainability: Why This Specific Prediction?

Logistic Regression: local contribution explanation
True=0, Predicted=0

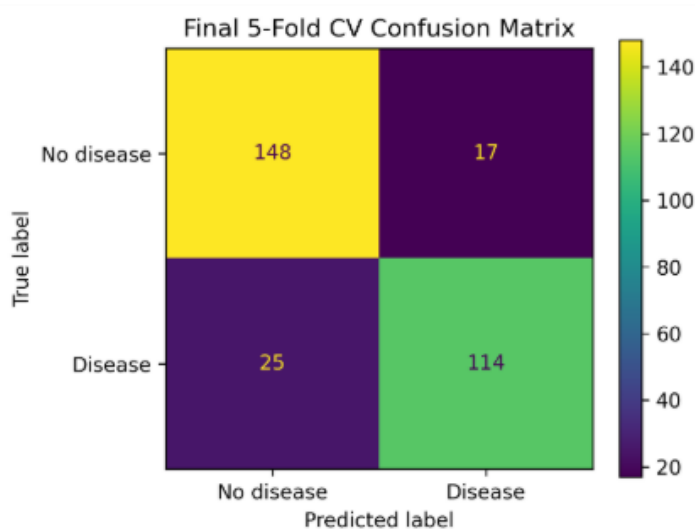
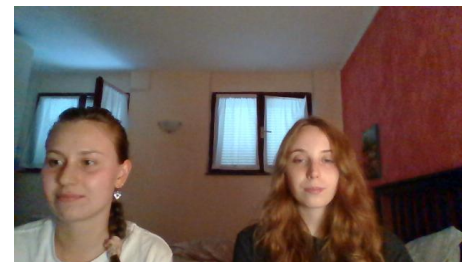


Case Analysis

- Patient prediction: This example shows how the model made a prediction for one individual patient.
- Main factors: The prediction was mainly influenced by variables such as ca, chest pain type, exercise-induced angina, and oldpeak. These features pushed the prediction toward the heart disease class.
- Consistency: The same variables were identified as important in the global explainability analysis. This indicates that the model uses similar information both globally and for individual predictions.
- The explanation is consistent with medical knowledge because these variables are known cardiovascular risk factors.
- This example shows a correctly classified patient.



Error Analysis: What Did the Model Get Wrong and Why?



Best-performing model: Logistic Regression on Cleveland unbalanced dataset

Reverse-Engineering Errors

- best-performing configuration: Logistic Regression trained on the unbalanced Cleveland dataset.
- A total of 42 samples were misclassified during 5-fold cross-validation. Most errors occurred in borderline cases where healthy and diseased patients showed similar clinical characteristics.
- The most important features: ca, chest pain type, exang, and oldpeak. Misclassifications likely occurred when these variables provided conflicting information.
- A false positive example showed that strong risk factors such as ca = 3 and asymptomatic chest pain pushed the prediction toward heart disease. At the same time, some features suggested a healthy patient, leading to an incorrect prediction.
- Heart disease diagnosis is complex, and some patients show characteristics of both classes.
- No clear systematic failure pattern was observed. Errors appeared to be spread across both classes.

Our ROC-AUC (0.917) is within the range commonly reported for the UCI Heart Disease dataset. Differences between studies may result from different features, preprocessing, validation strategies, and class balancing methods. The results support our main finding that feature quality was more important than dataset size.

Domain Impact, Limitations & Future Directions

Domain Impact

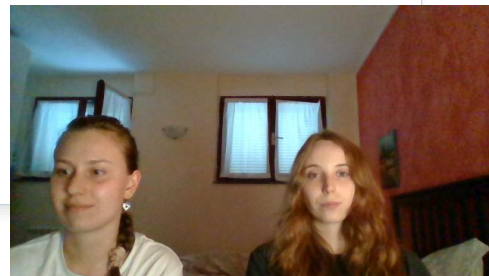
- The model could be used as a decision-support tool for early heart disease screening.
- It may help doctors identify high-risk patients and prioritize further examinations.
- False negatives are more critical than false positives because missed patients may not receive treatment in time.
- Final medical decisions should always be made by healthcare professionals.

Limitations

- The dataset is relatively small, especially the Cleveland subset.
- Only one dataset was used, which may limit generalization to other populations.
- Some clinically relevant features had to be removed from the Full dataset because of missing values.
- The study focused on model comparison rather than real-world deployment.

Future Directions

- Test the models on larger and more diverse datasets.
- Include additional clinical variables if available.
- Investigate more advanced ensemble and explainable AI methods.
- Evaluate performance in real clinical settings.



Thank You

Conclusions:

- We compared Logistic Regression, Decision Tree, and Random Forest for heart disease prediction using two dataset variants.
- The best model was Logistic Regression on the Cleveland dataset, evaluated using stratified 5-fold cross-validation.
- The most important predictors were ca, chest pain type, exercise-induced angina, and oldpeak, which are clinically meaningful risk factors.
- Feature quality was more important than dataset size. Future work should validate the models on larger and more diverse datasets.
- Dataset balancing does not improve results in lightly imbalanced datasets

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